

Computer Vision

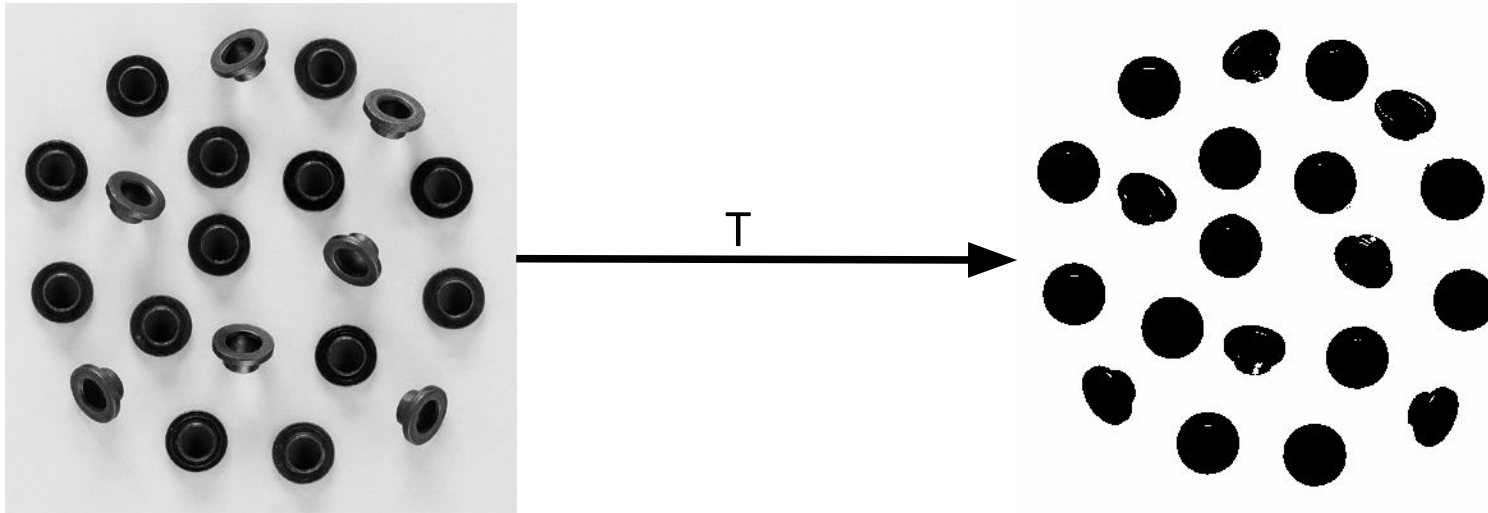
Gustavo Teodoro Laureano

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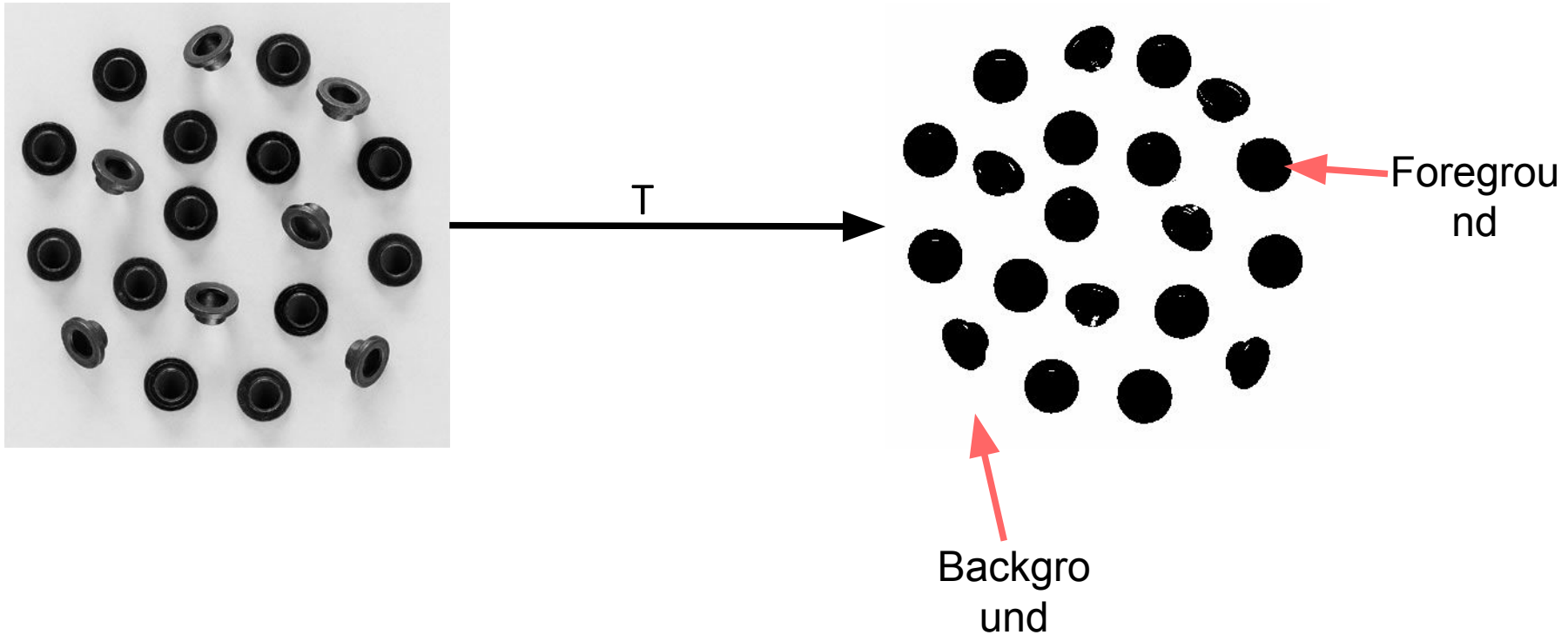
Programa de Pós-Graduação em Ciência da Computação
Instituto de Informática – Universidade Federal de Goiás

Image Binarization

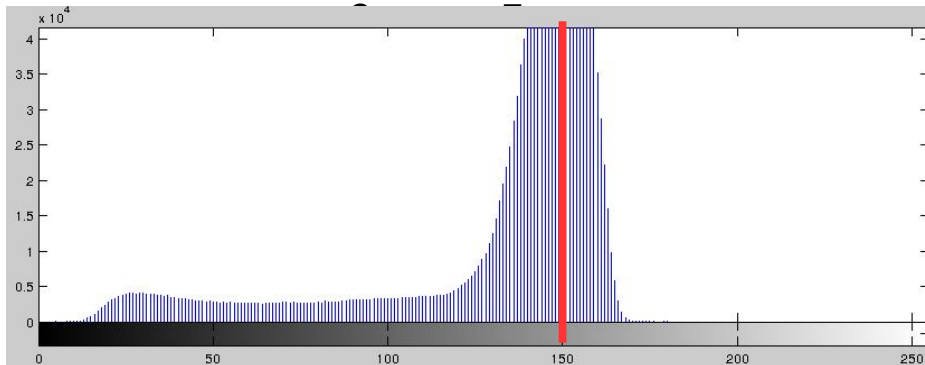
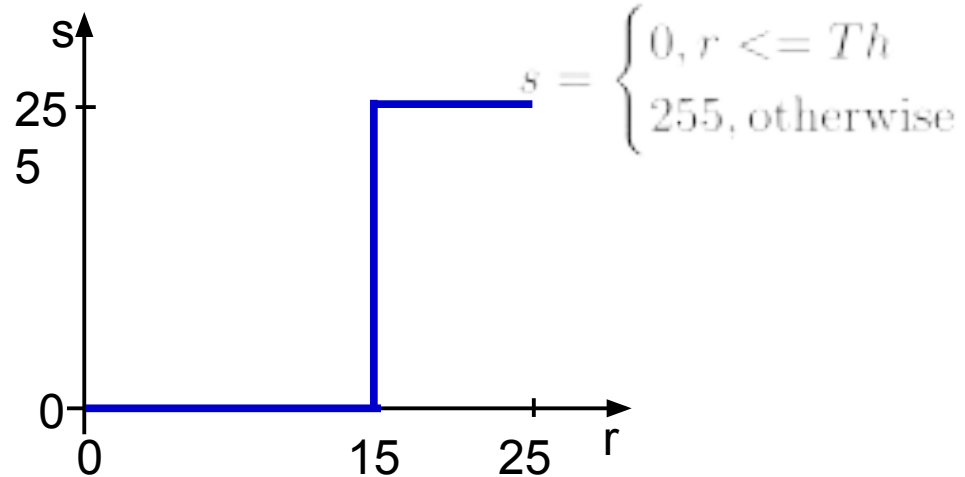
- Image Thresholding ...



- Image Thresholding ...



- Manual Thresholding

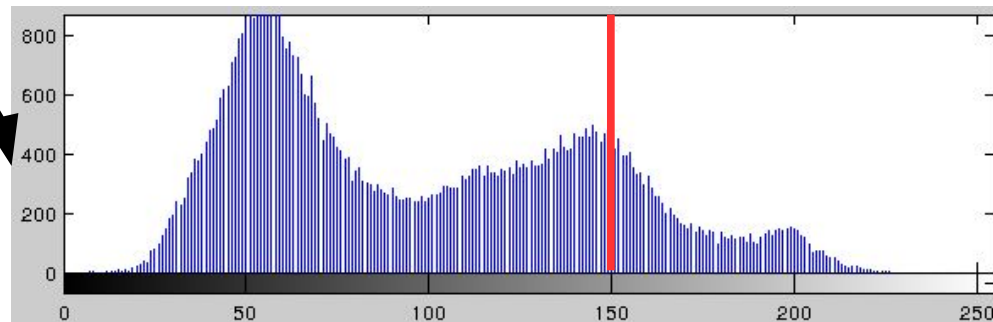
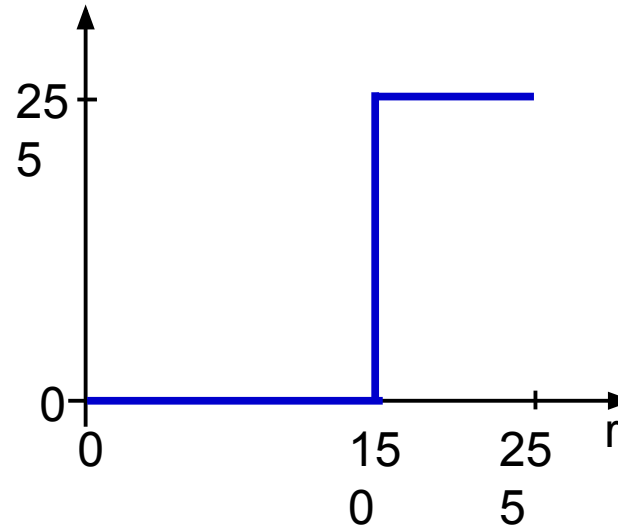


ponents or broken connection paths. There is no point past the level of detail required to identify those

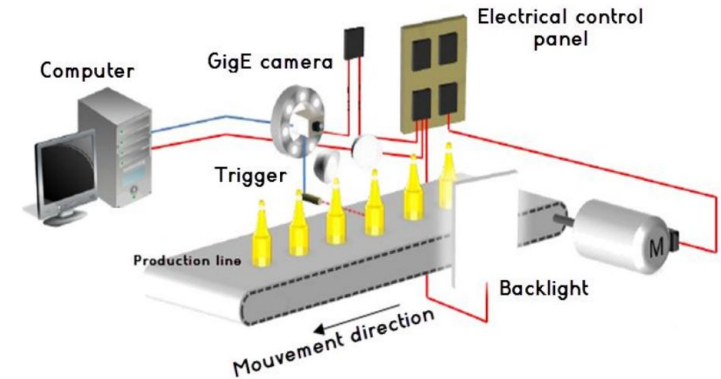
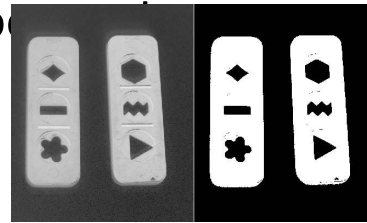
Segmentation of nontrivial images is one of the most difficult processing. Segmentation accuracy determines the evolution of computerized analysis procedures. For this reason, care must be taken to improve the probability of rugged segmentation, such as industrial inspection applications, at least some of the environment is possible at times. The experienced image designer invariably pays considerable attention to such

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•Manual Thresholding



- When use an empirical threshold value?
- In controlled conditions, an empirical threshold may be usefully.
- Illumination, Noise, Background must be known.
- When the two distributions are reasonably well separated:
- some overlap between classes but not too much
- Examples:
- Industry
- Assembly-line
- Visual Inspection
- Off-the-shelf products



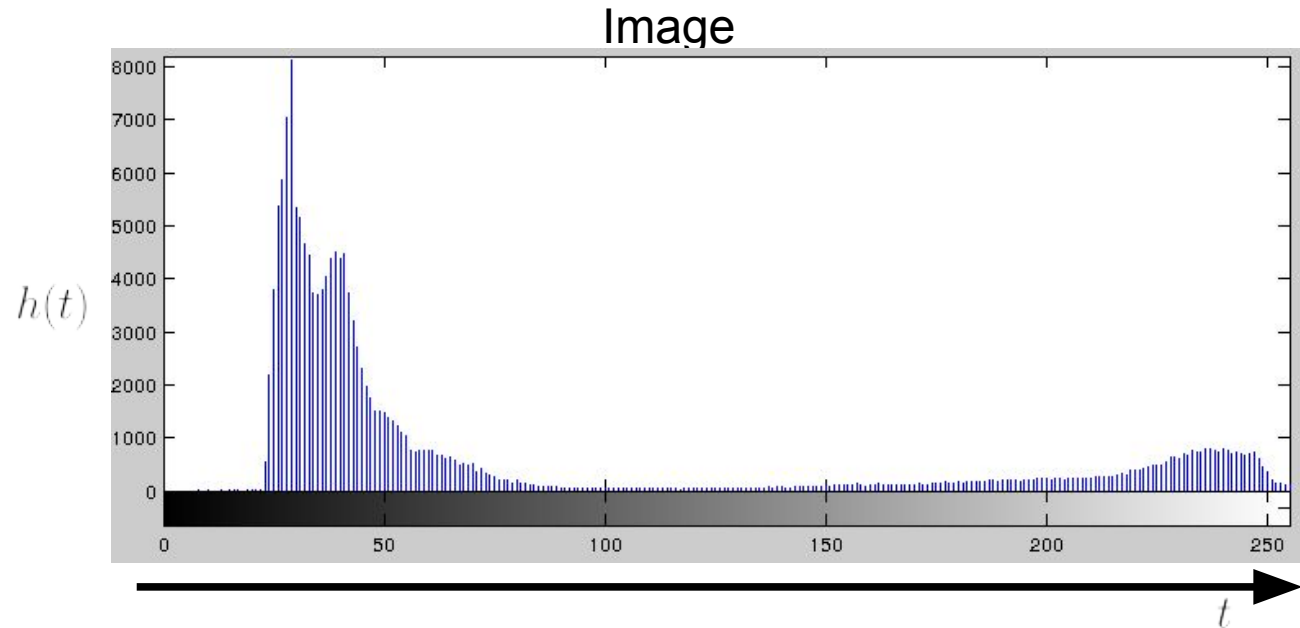
- Xerox machines

- Adaptive Thresholding:
 - Global Methods
 - Local Methods
 - In both there is the question: **What the Best Threshold Value?**
-

•What the Best Threshold Value for this image?



Grayscale
Image

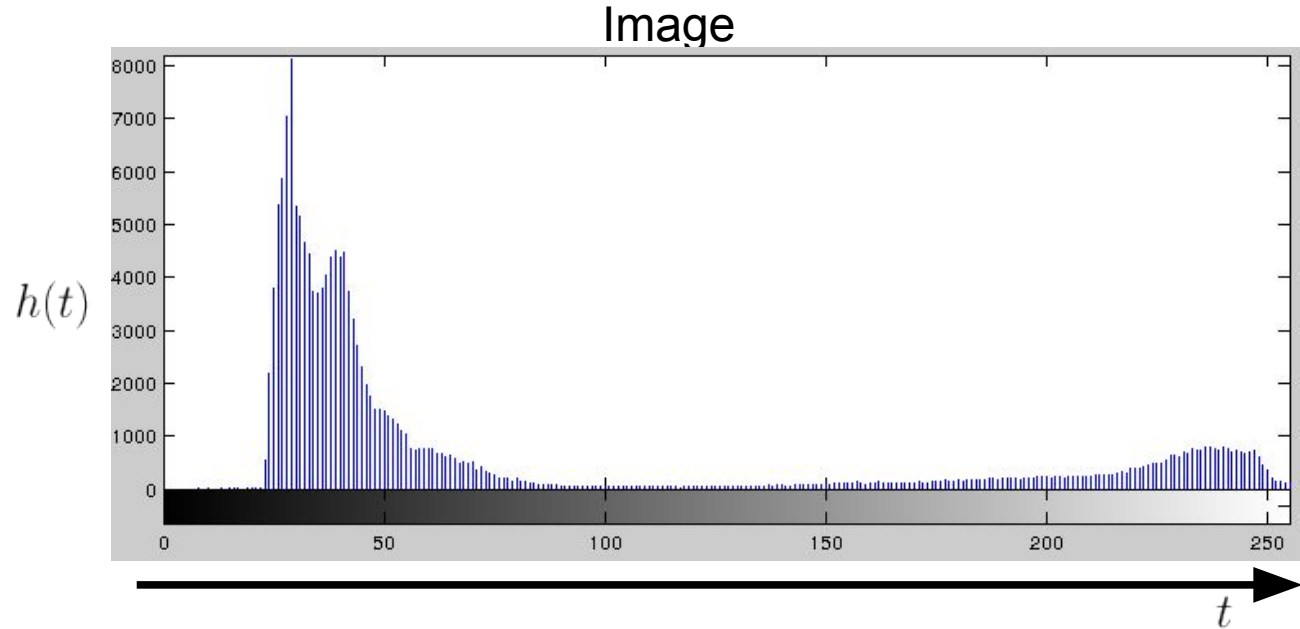


- How to compute histogram?

```
function [ h ] = histogram( I )  
    h = zeros(1,256);  
    for i = 1 : length( I(:) )  
        h( I(i)+1 ) = h( I(i)+1 ) + 1;  
    end  
end
```

or

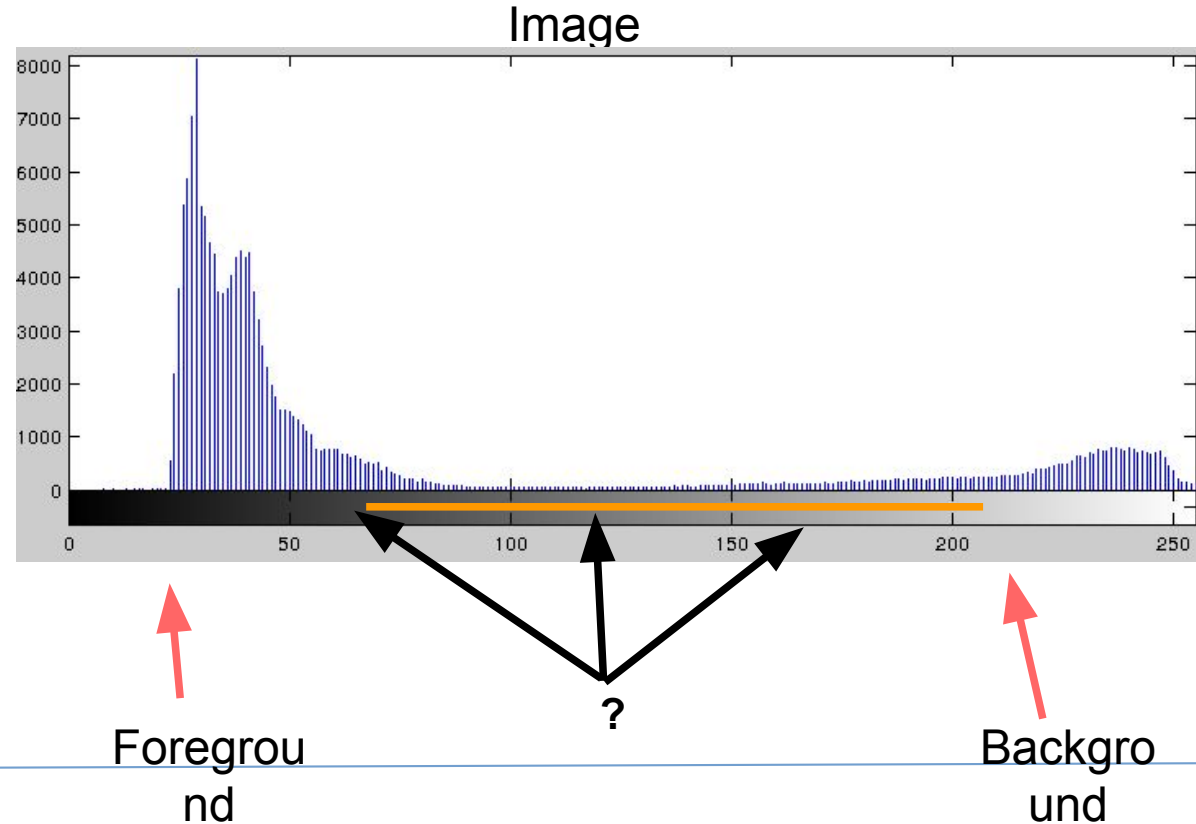
```
h = imhist( I );
```



- What the Best Threshold Value for this image?



Grayscale
Image



•What the Best Threshold Value for this image?

–Otsu 's Method

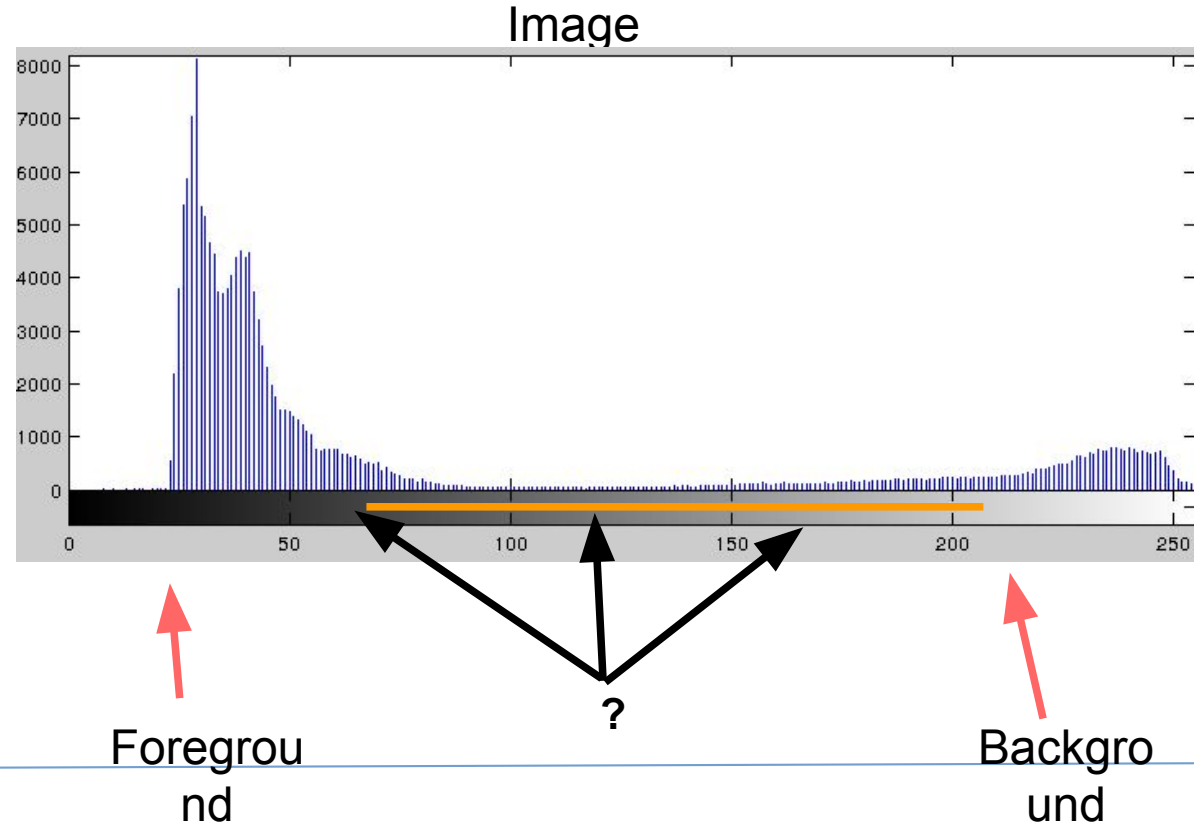
A Threshold Selection Method from Gray-Level Histograms

NOBUYUKI OTSU

Abstract—A nonparametric and unsupervised method of automatic threshold selection for picture segmentation is presented. An optimal threshold is selected by the discriminant criterion, namely, so as to maximize the separability of the resultant classes in gray levels. The procedure is very simple, utilizing only the zeroth- and the first-order cumulative moments of the gray-level histogram. It is straightforward to extend the method to multithreshold problems. Several experimental results are also presented to support the validity of the method.

I. INTRODUCTION

It is important in picture processing to select an adequate threshold of gray level for extracting objects from their background. A variety of techniques have been proposed in this regard. In an ideal case, the histogram has a deep and sharp valley between two peaks representing objects and background, respectively, so that the threshold can be chosen at the bottom of this valley [1]. However, for most real pictures, it is often difficult to detect the valley bottom precisely, especially in such cases as when the valley



OTSU, N. 1979. *A Threshold Selection Method from Gray-Level Histograms*

• What the Best Threshold Value for this image?

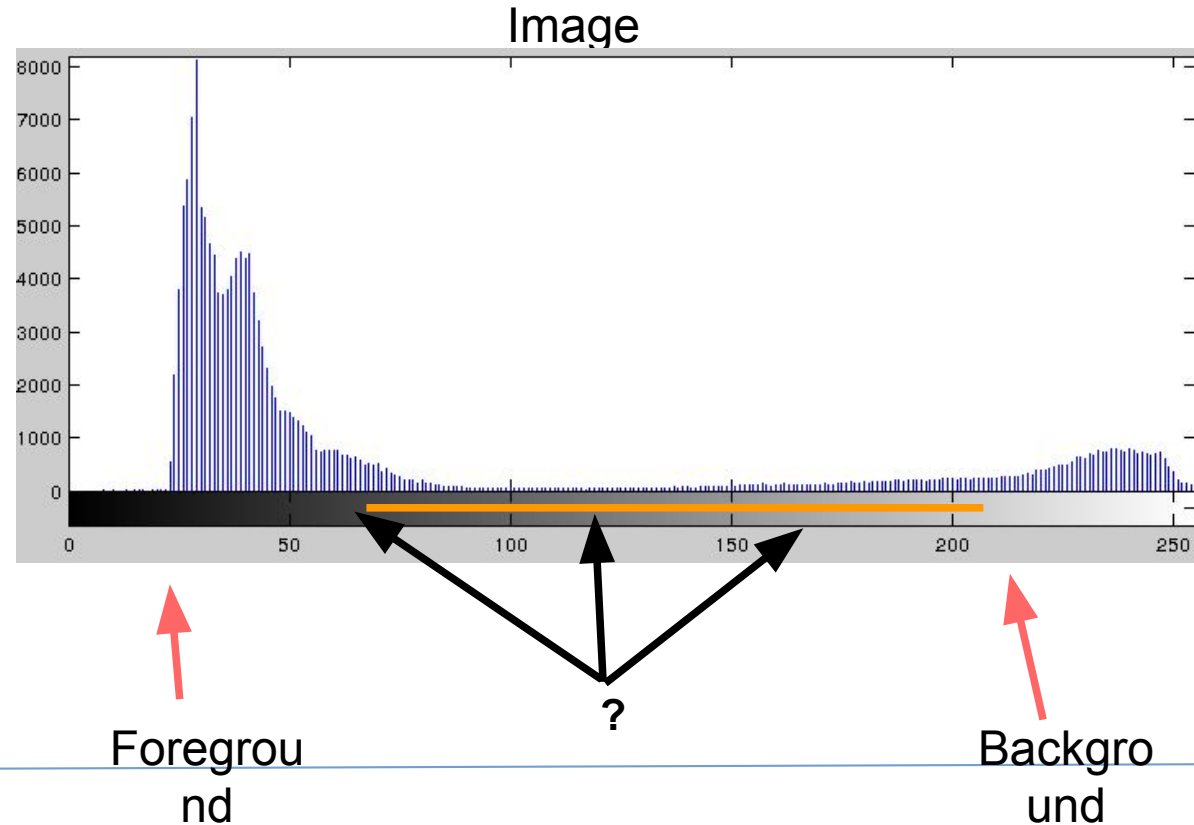
– Otsu 's Method

– Exhaustive search for Th that:

•

• or

•



• What the Best Threshold Value for this image?

– Otsu 's Method

– Exhaustive search for Th that:

•

• or

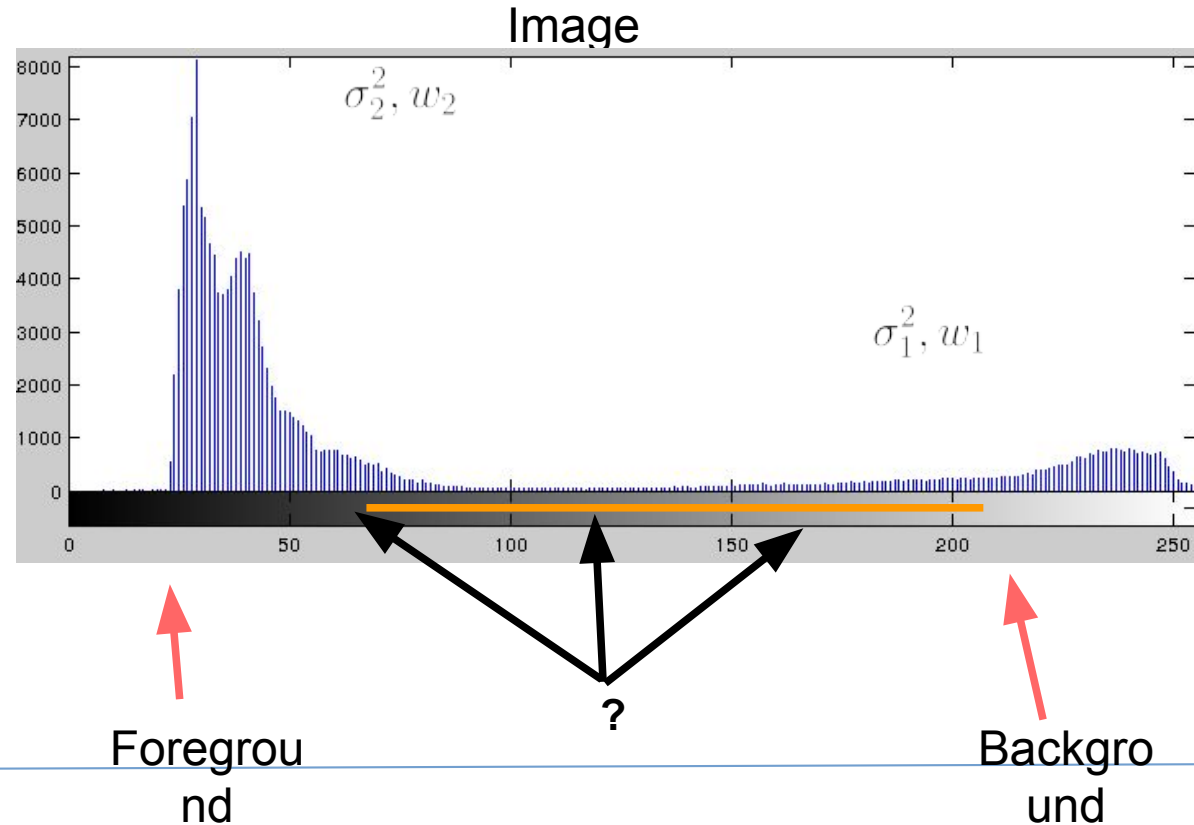
•

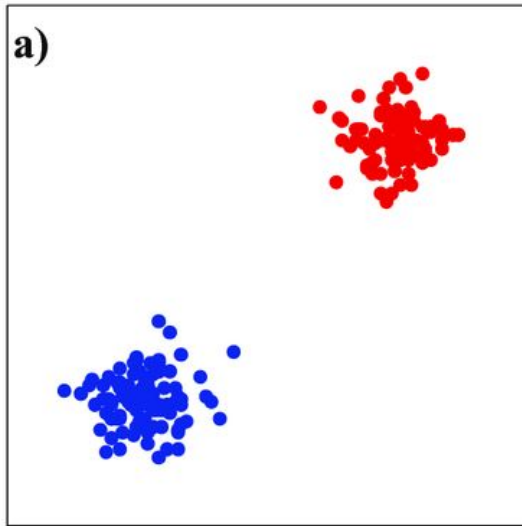
σ_1^2 : background variance

σ_2^2 : foreground variance

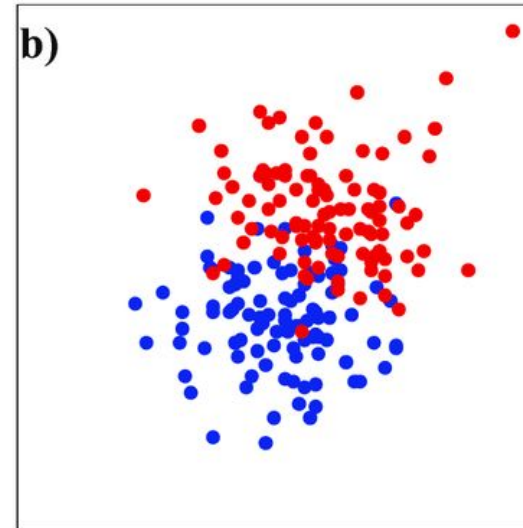
w_1 : background weight

w_2 : foreground weight





a) Low intra-class variance and high inter-class variance: compact well separated clusters



b) High intra-class variance and low inter-class variance: wide clusters without a clear frontier.

•What the Best Threshold Value for this image?

–Intra-class variance (within):

$$\sigma_w^2(t) = w_1(t)\sigma_1^2(t) + w_2(t)\sigma_2^2(t)$$

$\sigma_1^2(t)$: background variance

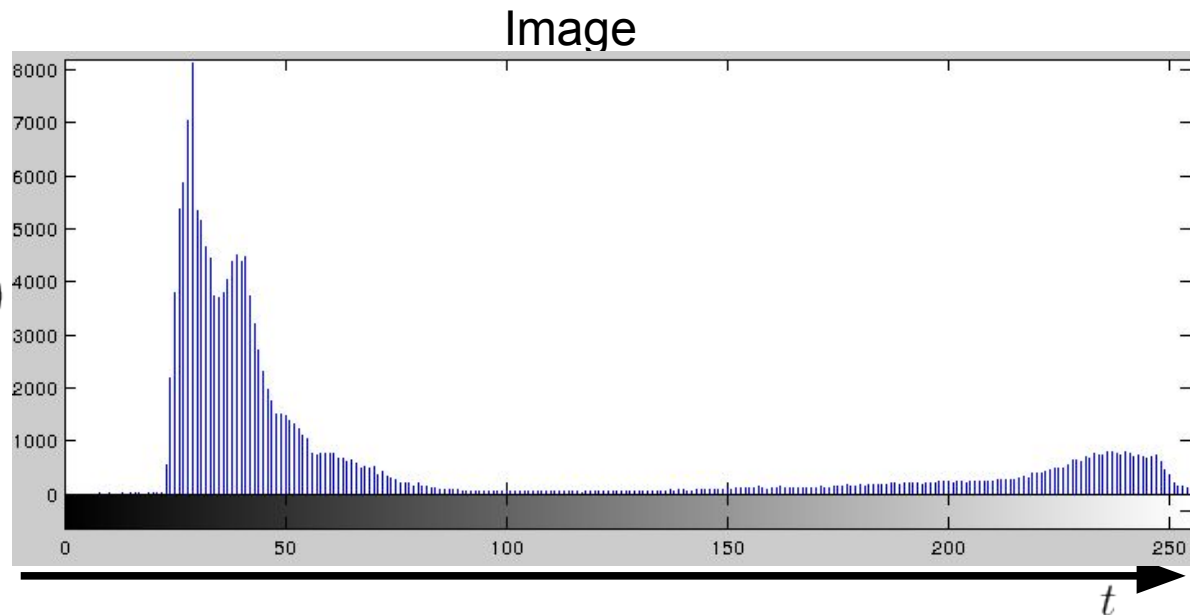
$\sigma_2^2(t)$: foreground variance

$w_1(t)$: background weight $h(t)$

$w_2(t)$: foreground weight

–Otsu 's Method:

$$\arg \min_t \sigma_w^2(t)$$



- What the Best Threshold Value for this image?

- Intra-class variance (within):

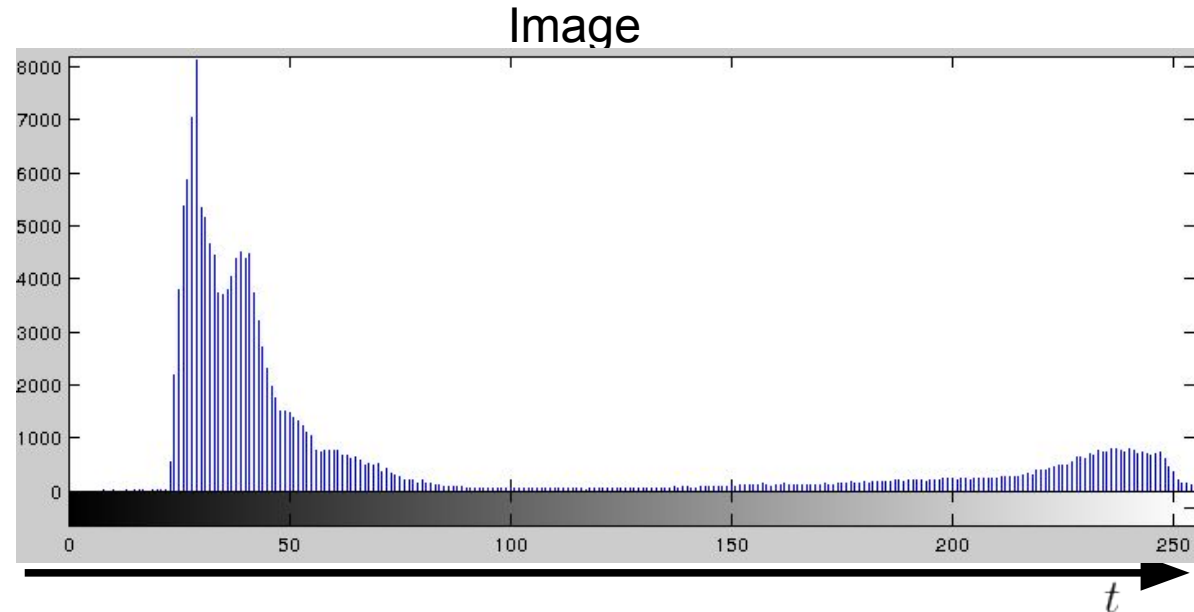
$$\sigma_w^2(t) = w_1(t)\sigma_1^2(t) + w_2(t)\sigma_2^2(t)$$

- Inter-class variance (between):

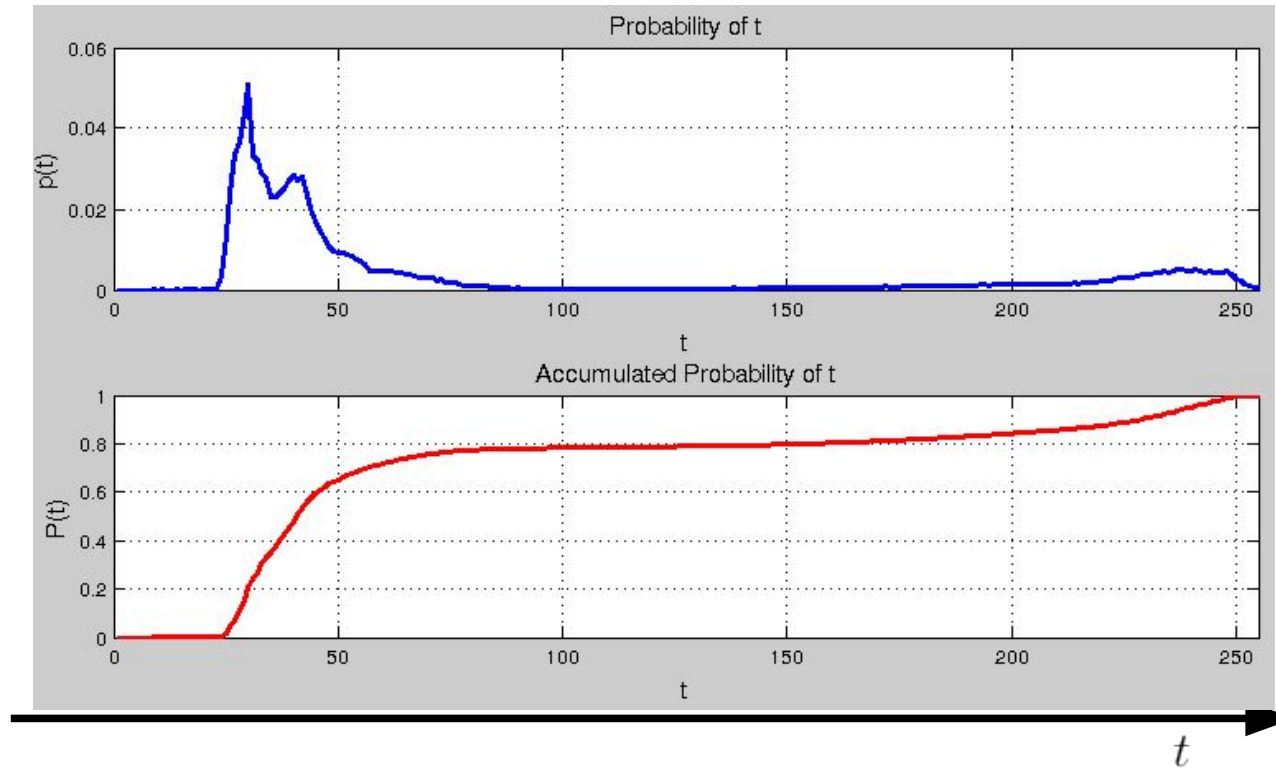
$$\sigma^2 = \sigma_w^2 + \sigma_b^2$$

- Otsu 's Method:

$$\arg \min_t \sigma_w^2(t) \Leftrightarrow \arg \max_t \sigma_b^2(t)$$



•Otsu 's Method:



- What the Best Threshold Value for this image?

- Otsu's Algorithm $\arg \min_t \sigma_w^2(t) \Leftrightarrow \arg \max_t \sigma_b^2(t)$

$$\begin{aligned}\sigma_b^2 &= \sigma^2 - \sigma_w^2 \\ &= w_1(\mu_1 - \mu)^2 + w_2(\mu_2 - \mu)^2 \\ &= w_1 w_2 (\mu_1 - \mu_2)^2\end{aligned}$$

where $\mu = w_1 \mu_1 + w_2 \mu_2$

- Probability of t and Acc.: $p(t) = \frac{h(t)}{WH}$ $P(t) = \sum_{i=0}^t p(i)$

- Weights: $w_1(t) = \sum_{i=0}^t p(i) = P(t)$
 $w_2(t) = \sum_{i=t+1}^{L-1} p(i) = P(L-1) - P(t)$
- Accumulated Probability

- Otsu 's Method:

// Total number of pixels

int total = W*H;

// Calculate histogram

for (int t=0 ; t< 256 ; t++) hist[t]=0;

for (int i=0 ; i< total ; i++) hist[data[i]]++;

// Calculate total sum

float sum = 0;

for (int t=0 ; t<256 ; t++) sum += t * hist[t];

float sumB = 0;

int w1 = 0;

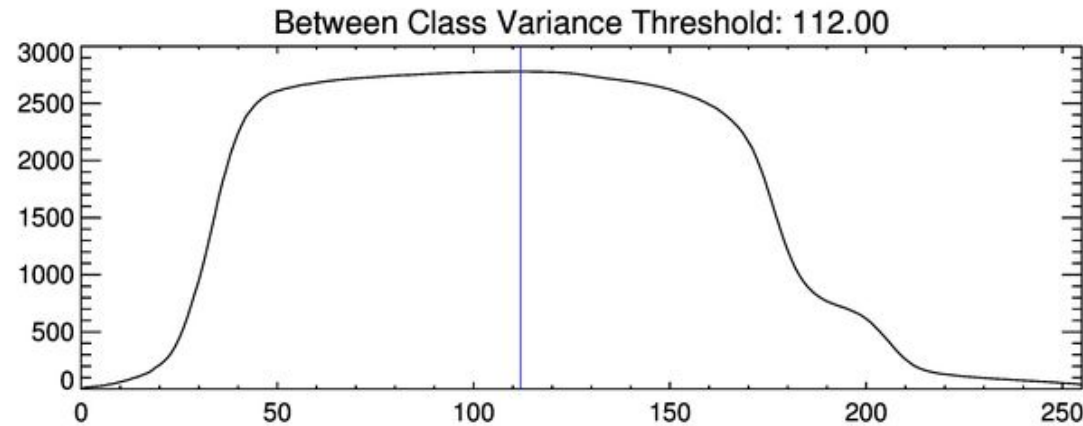
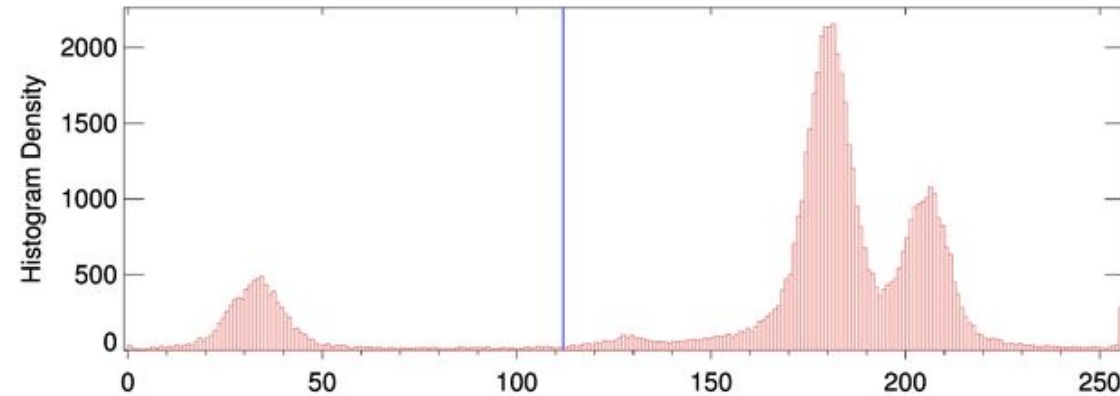
int w2 = 0;

float varMax = 0;

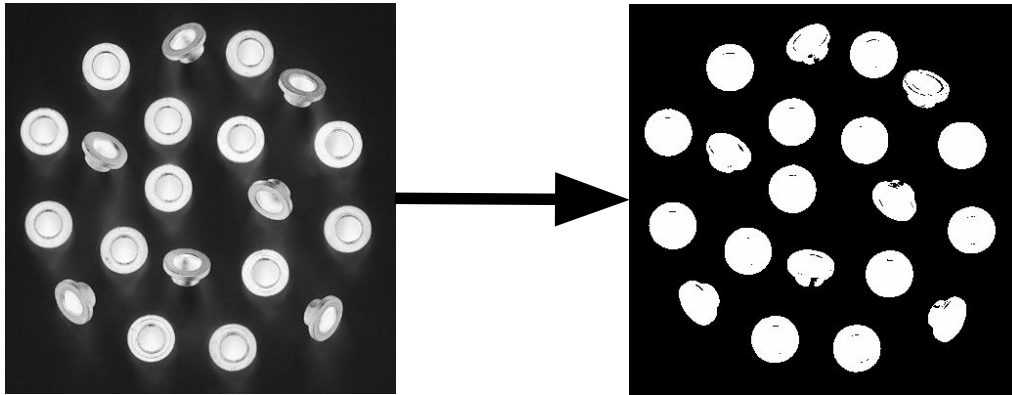
char threshold = 0;

```
for (int t=0 ; t<256 ; t++) {  
    w1 += histData[t];           // Weight Background  
    if (w1 == 0) continue;  
    w2 = total - w1;           // Weight Foreground  
    if (w2 == 0) break;  
    sum1 += (float) (t * histData[t]);  
    float m1 = sum1 / w1;       // Mean Background  
    float m2 = (sum - sum1) / w2; // Mean Foreground  
    // Calculate Between Class Variance  
    float varBetween = (float)w1 * (float)w2 * (m1 - m2) * (m1 -  
m2);  
    // Check if new maximum found  
    if (varBetween > varMax) {  
        varMax = varBetween;  
        threshold = t;  
    }  
}
```

- Otsu 's Method:



- Otsu's Method:



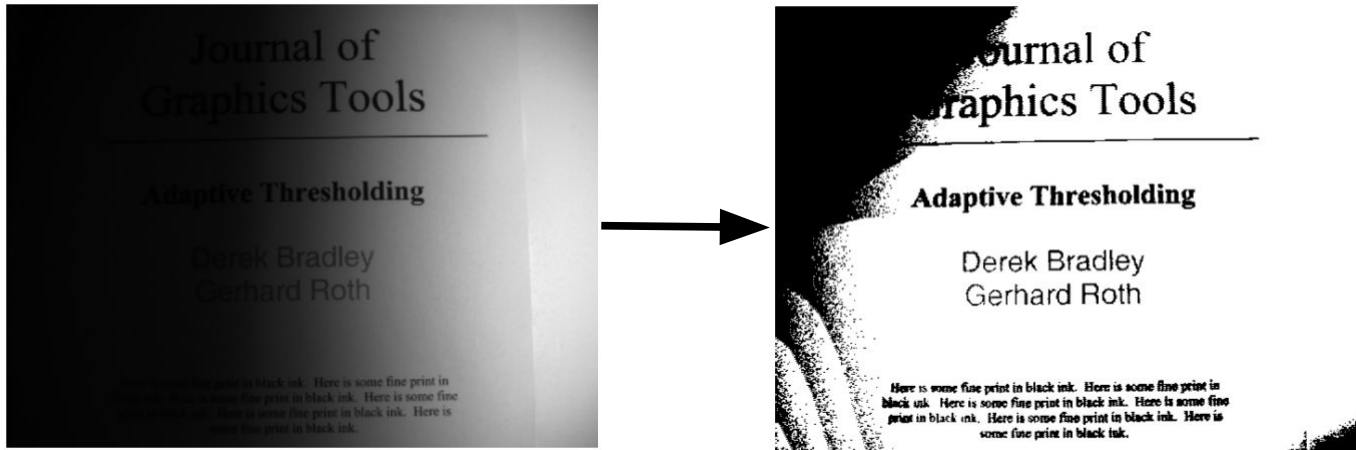
ponents or broken connection paths. There is no point in going past the level of detail required to identify those components.

Segmentation of nontrivial images is one of the most difficult tasks in image processing. Segmentation accuracy determines the effectiveness of computerized analysis procedures. For this reason, considerable effort can be taken to improve the probability of rugged segmentation, especially in applications such as industrial inspection applications, at least some of the time. The experienced designer invariably pays considerable attention to such

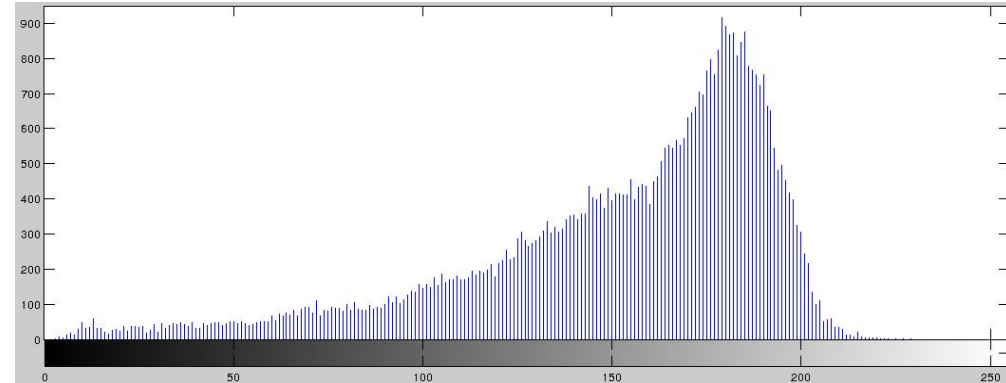
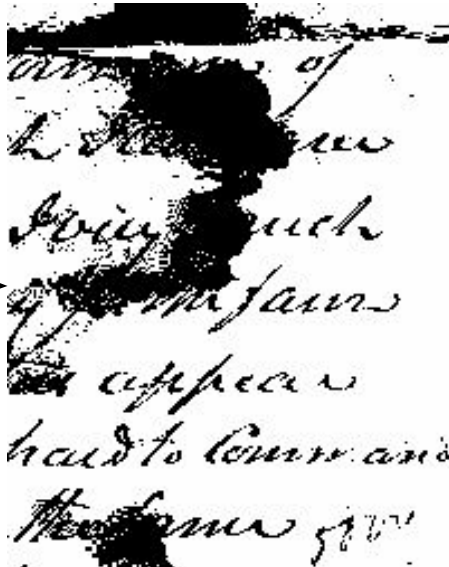
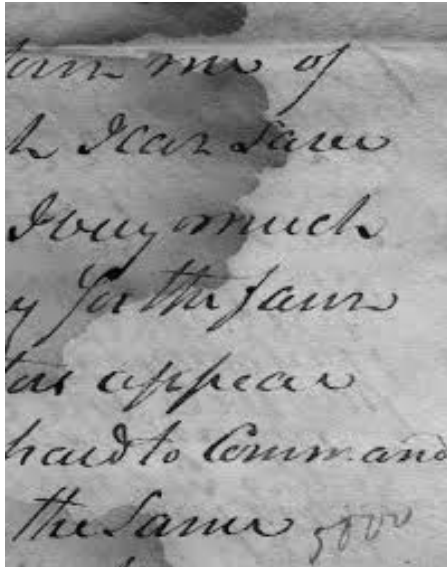
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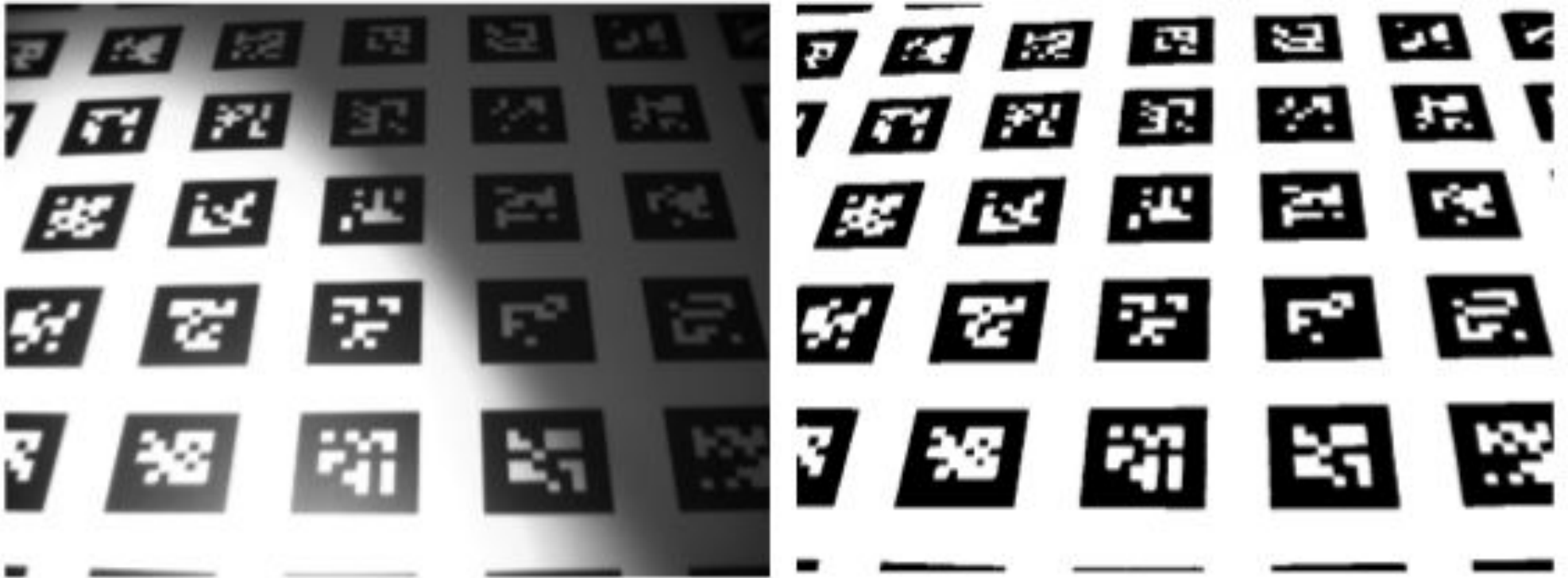
- Otsu's Method:



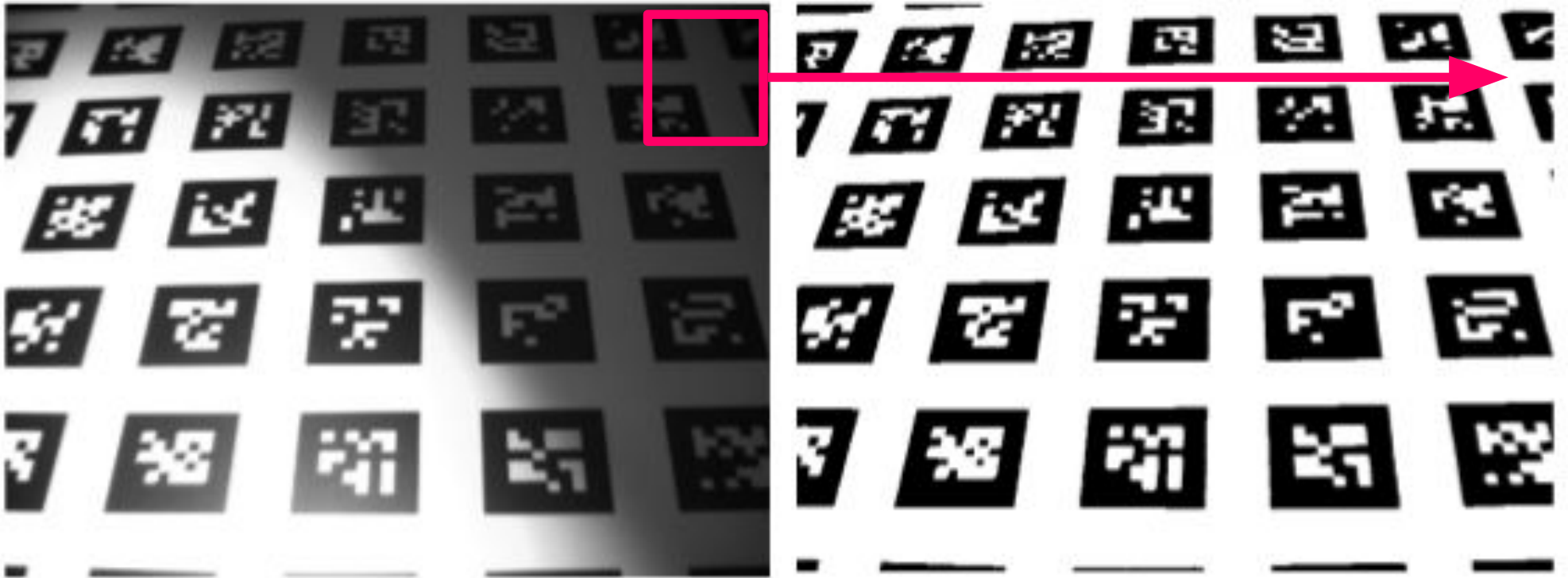
- Otsu's Method:



- Local Methods



- Local Methods



- Local Methods

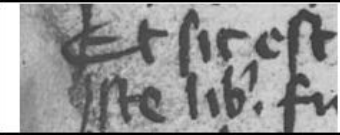
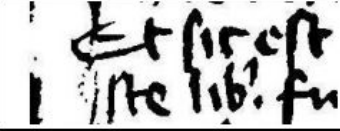
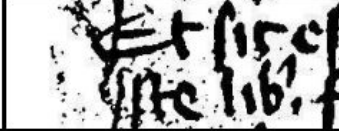
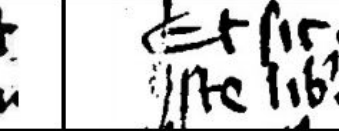
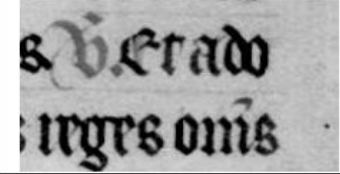
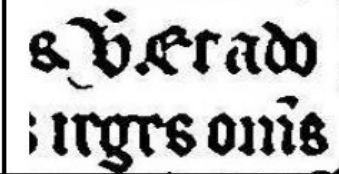
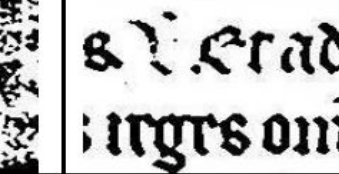
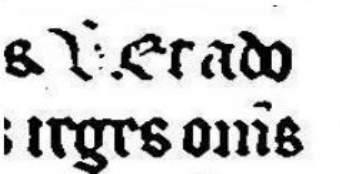
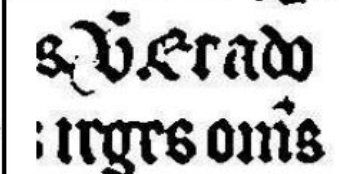
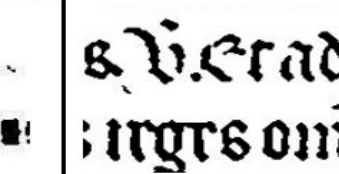
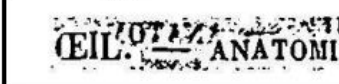
$$\arg \min_t \sigma_w^2(t) \Leftrightarrow \arg \max_t \sigma_b^2(t)$$

$$T_{Niblack} = \mu + k\sigma$$

$$T_{Sauvola} = \mu(1 - k(1 - \frac{\sigma}{R}))$$

- How to measure accuracy of thresholding methods?

- How to measure accuracy of thresholding methods?
 - Visual

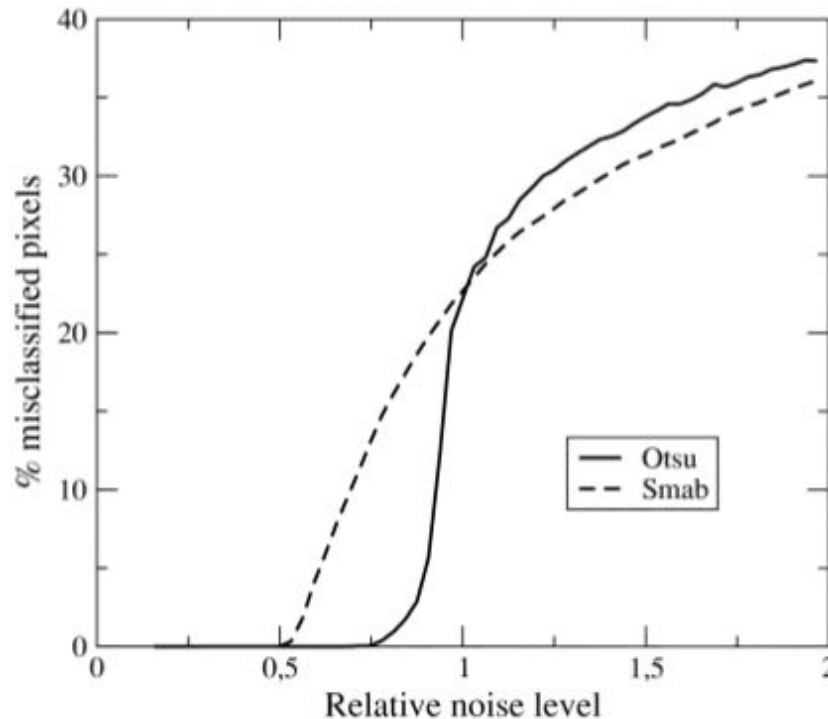
		
		
		
		
		
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- How to measure accuracy of thresholding methods?
 - Influence in final results

Image	Total Characters	Number of characters correctly recognized				
		Niblack	Sauvola	Wolf	Feng	NICK
Image 1	2012	0	1907	2003	1932	2002
Image 2	944	887	0	821	938	940
Image 3	241	239	233	233	230	238
Image 4	364	0	356	354	344	363
Other 21 images	5446	2687	3783	4828	5364	5411
Total	9007	3813	6279	8239	8808	8954
RECG RATE		42.33	69.71	91.47	97.79	99.41

- How to measure accuracy of thresholding methods?
 - Using a ground truth image

- How to measure accuracy of thresholding methods?
 - Counting misclassified points after inject noise



Torbjørn Sund and Karsten Eilertsen (2003).
An Algorithm for Fast Adaptive Image Binarization With Applications in Radiotherapy Imaging